

Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP395B007-V4

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1. Introduction

1.1 Scope of application

This specification applies to the LCD module that is supplied by Shenzhen Osprey Optoelectronics Technology Co., Ltd.

LCD specification: Dots 480xRGBx480.

As to basic specification of the driver IC, refer to the IC (ST7102) specification and data book.

All material & processing of the LCD module should be Lead Free.

1.2 TFT features:

Structure: TFT PANNEL+IC +FPC+BL;

ALL Viewing Type LCD

480dot-segment and 480 dot-common outputs;

16.7M Color can be selected by software;

White LED back light;

3SPI +18RGB interface

The logo for Osptek, featuring a stylized bird or wing graphic in light blue and purple above the word "osptek" in a lowercase, sans-serif font, with a registered trademark symbol (®) to the upper right of the "k".

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2. LCM General specification

ITEM	Standard value	Unit
LCD Type	Normally black	--
Drive element	TFT active matrix	--
Number of pixels	480*3RGB(H)X480(V)	Dots
Pixel arrangement	RGB stripe	--
Pixel Pitch (W*H)	0.1497x0.1462	mm
Active area	71.86(H) x 70.18(V)	mm
Viewing direction	ALL O'CLOCK	--
TFT Driver IC	ST7102	--
TFT interface	MIPI 2L interface	--
Approx. Weight	TBD	g
LCM Size(W*H*T)	74.66(H) x 76.54(V) x2.06(T)	mm
Touch structure	--	--
Touch Driver IC	--	--
Touch Interface	--	--

3.Absolute Maximum Rating

Characteristics	Symbol	Min.	Max.	Unit
LCM Operating Temperature	T _{OPR}	-20	+70	°C
LCM Storage Temperature	T _{STG}	-30	+80	°C
Humidity	RH	-	90	%

4.Electrical Characteristics

4.1 TFT DC Characteristics

Characteristics	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage for(DC/DC)	VCI	2.8	2.8	3.3	V

4.2 Back-Light Unit Characteristics

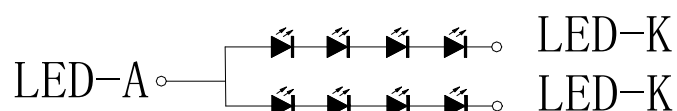
The back-light system is an edge-lighting type with 8 white LEDs. The characteristics of the back-light are shown in the following tables.

Characteristics	Symbol	Min.	Type	Max.	Unit	Notes
Forward Voltage	V _F	11.2	--	12.8	V	-
Forward current	I _F	--	40	--	mA	-
Luminance(With LCD)	L _v	--	350	--	cd/m ²	-
LED life time	N/A	--	30,000	--	Hr	Note 1

Note:

- (1) The “LED life time” is defined as the module brightness decrease to 50% of original brightness at I_L=20mA/LED. The LED life time could be decreased if operating I_L is larger than 25mA/LED.

Backlight circuit diagram shown in below:



5. Module Function Description

Pin No.	Symbol	Functional	Notes
1	GND	Power Ground	
2	D0_N	MIPI DSI differential data pair	
3	D0_P	MIPI DSI differential data pair	
4	GND	Power Ground	
5	D1_N	MIPI DSI differential data pair	
6	D1_P	MIPI DSI differential data pair	
7	GND	Power Ground	
8	CLK_N	MIPI DSI differential clock pair	
9	CLK_P	MIPI DSI differential clock pair	
10	GND	Power Ground	
11	NC	No Connection.	
12	NC	No Connection.	
13	GND	Power Ground	
14	NC	No Connection.	
15	NC	No Connection.	
16	GND	Power Ground	
17	NC	No Connection.	
18	NC	No Connection.	
19	GND	Power Ground	
20	TP_SCL 1.8V	Touch panel I2C clock	
21	TP_SDA 1.8V	Touch panel I2C data	
22	VCC 3.3V	Analog supply voltage	
23	VCC 3.3V	Analog supply voltage	
24	TE	Tearing effect signal is used to synchronize MCU to frame memory	
25	BL+5V	POWER SUPPLY FOR BLU	
26	BL+5V	POWER SUPPLY FOR BLU	
27	TP_INT 1.8V	Touch panel interrupt output	
28	TP_RST 1.8V	Touch panel reset	
29	RESET 1.8V	LCM Reset signal.	
30	BL_PWM	PWM dimming control input	

6. Timing Characteristics

The power On/Off timing diagram is illustrated below.

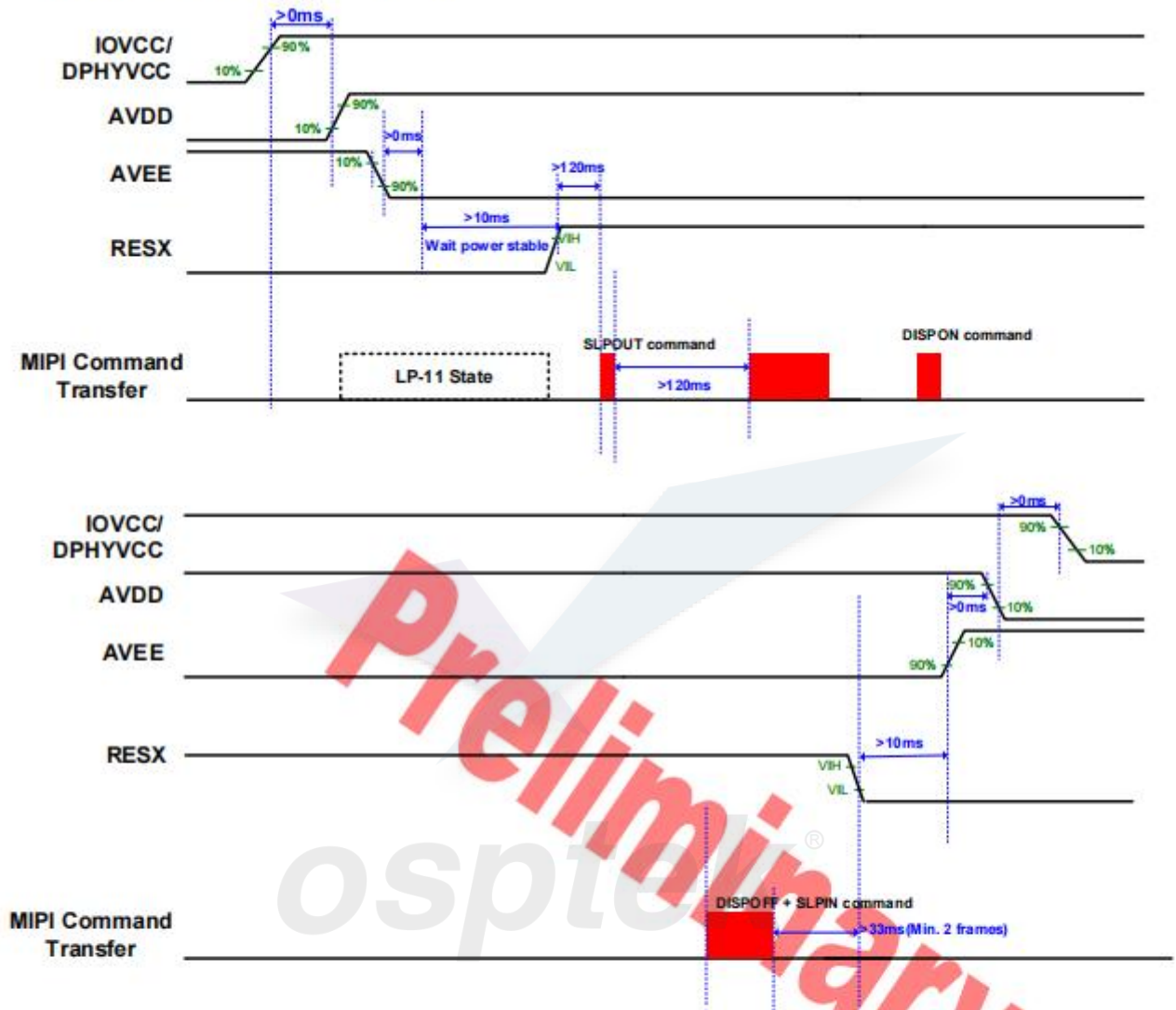


Figure 72 The power on sequence timing

Note 1: MIPI lanes must go to LP11 after Power IOVCC/DPHYVCC is ready

Note 2: After SLPOUT command, driver IC will start internal power on action. Any other settings should be set after SLPOUT command with a minimum of 120mS.

Note 3: DISPOFF and SLPIN command should be set after SLPOUT command with a minimum delay time of 120mS.

Note 4: RESX tied low into deep standby mode with a minimum time of 10ms.

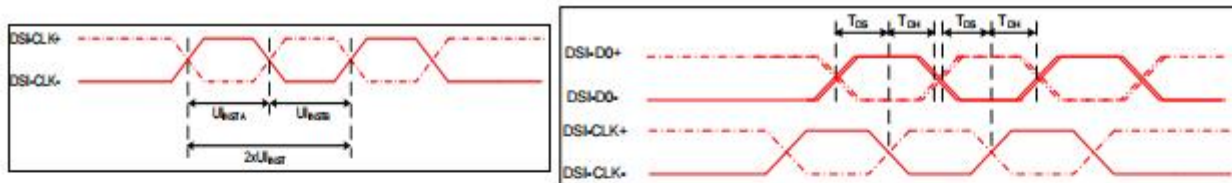
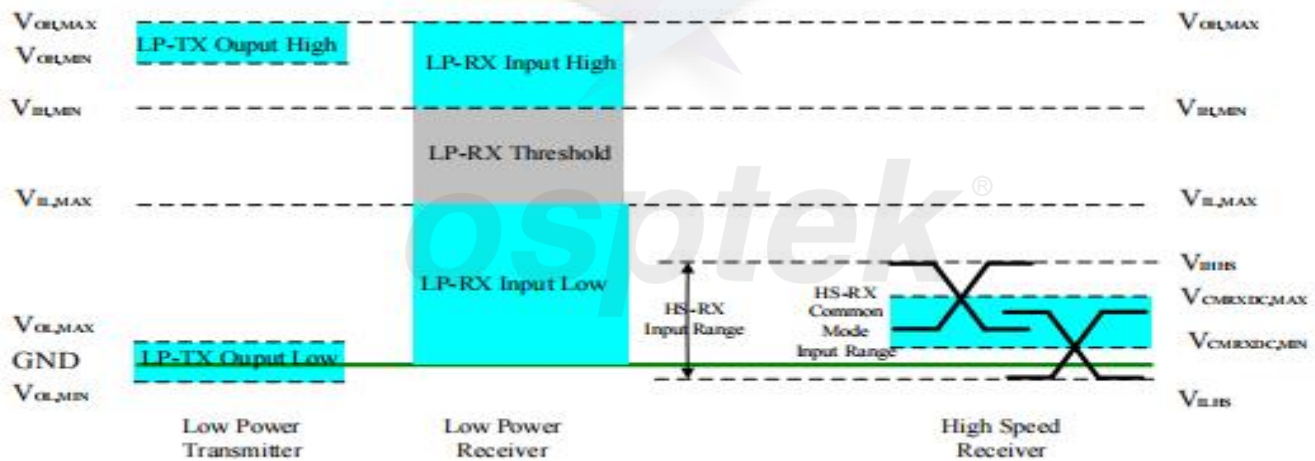


Figure 77 High Speed Mode – Clock Channel Timing

Signal	Symbol	Parameter	Specification			Unit	Description
			MIN	TYP	MAX		
DSI-CLK+/-	2xUINST	Double UI instantaneous	1.6	-	10	ns	
DSI-CLK+/-	UINSTA, UINSTB	UI instantaneous half	0.8	-	5	ns	
fDSI-CLK+/-	FDSCLK	DSI-CLK+/- frequency	100	-	625	MHz	
DSI-Dn+/-	TDS	Data to clock setup time	0.15	-	-	UI	
DSI-Dn+/-	TDH	Data to clock hold time	0.15	-	-	UI	



Reset function

The Reset function of ST7102 is triggered by RESX input. After reset function is triggered, the ST7102 is into a reset period, and the duration of this period must be at least 1ms. During this period, the ST7102 and its power circuit is initialized.

Initial State Of Output Pins

Output Pins Name	Initial State
S[480:1] (source output)	GND
VGH	GND
VGL	GND
VCOM	GND
CGOUTL_R[16:1](GIP signal)	GND

Reset Timing Diagram

ST7102 provides Power On Reset and HWRST pin (RESX) for IC initialization and exit Deep Standby Mode (DSTB). For exiting DSTB Mode, HWRST pin should be tied to high at least 10ms. The timing diagram is located at below.

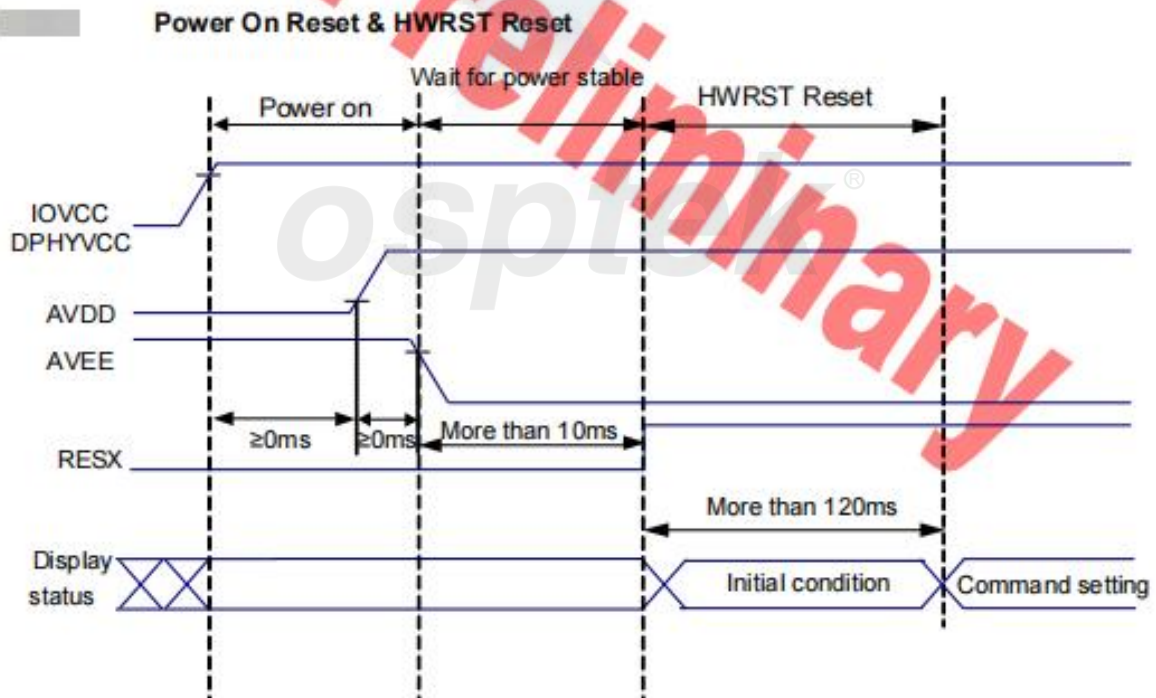


Figure 67 Power on Reset and HWRST reset

Note: RESX should be tied from low to high when exiting DSTB Mode.

7.Optical Characteristics

<Table 5. Optical Specifications>

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle range	Horizontal	Θ_3	CR > 10	80	85	-	Deg.	Note 1
		Θ_9		80	85	-	Deg.	
	Vertical	Θ_{12}		80	85	-	Deg.	
		Θ_6		80	85	-	Deg.	
Luminance Contrast ratio		CR	$\Theta = 0^\circ$	900	1200	-		Note 2
Cell Transmittance		Tr		4.0	4.5		%	@Silicate
Color Gamut	NTSC	CIE1931	$\Theta = 0^\circ$	65	70	-	%	@C Light
Reproduction of Color	Red	Rx	$\Theta = 0^\circ$	Typ-0.03	0.656	Typ+0.03		
		Ry			0.339			
	Green	Gx			0.286			
		Gy			0.611			
	Blue	Bx			0.138			
		By			0.098			
	White	Wx			0.307			
		Wy			0.318			
Response Time		Tr+Td	Ta= RT $\Theta = 0^\circ$	-	30	35	ms	Note 4

Parameter	Condition	Min.	Typ.	Max.	Remark
Flicker		-	-	-30dB	@灰黑 Pattern L127灰阶
Gamma		-0.3	2.2	+0.3	@ Ave
Crosstalk		-	-	2%	25°C

Notes : 1. Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing angles are determined for the horizontal or 3, 9 o'clock direction and the vertical or 6, 12 o'clock direction with respect to the optical axis which is normal to the LCD surface (see FIGURE 1).

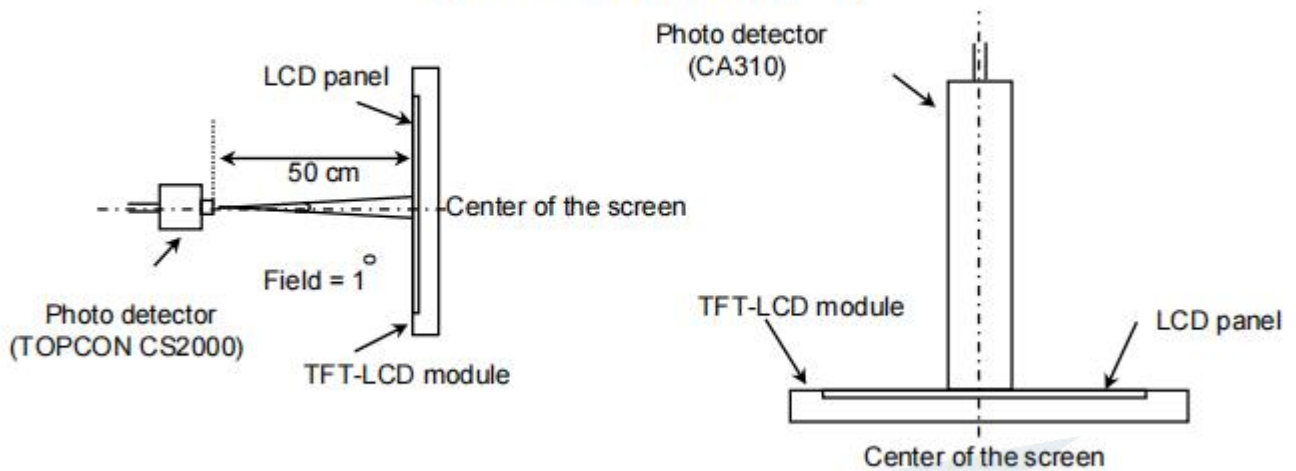
2. Contrast measurements shall be made at viewing angle of $\Theta = 0$ and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state . (see FIGURE 1) Luminance Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

3. The color chromaticity coordinates specified in Table 5. shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.
4. The electro-optical response time measurements shall be made as FIGURE 2 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is T_r , and 90% to 10% is T_d .

Optical measurements

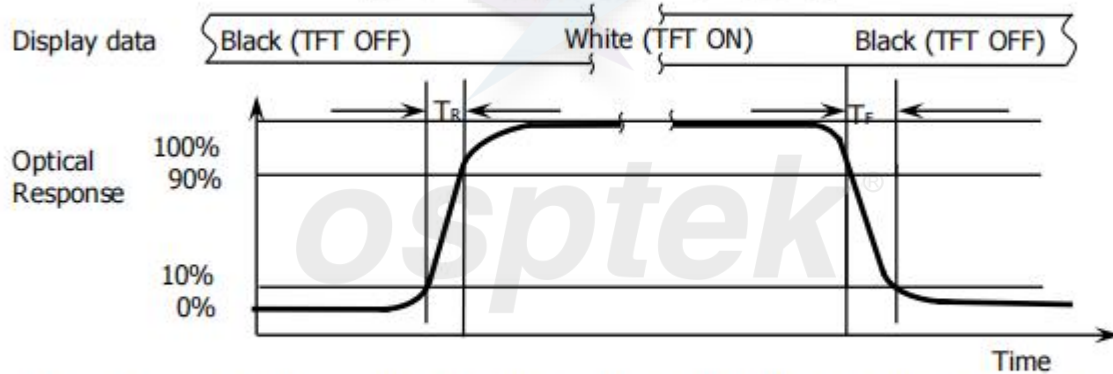
Figure 1. Measurement Set Up



View angel range, uniformity, etc. measurement setup

Flicker, measurement setup

Figure 2. Response Time Testing



The electro-optical response time measurements shall be made as shown in FIGURE 3 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is T_r and 90% to 10% is T_d .

8. Reliability Test Item

No.	Test Item	Test Condition	Notes
1	High Temp. Storage	+80°C / 96H	1. Functional test is OK. Missing Segment, short, unclear segment non-display, display abnormally and liquid crystal leakage un-allowed. 2. No low temperature bubbles, end seal loose and fall, frame rainbow.
2	Low Temp. Storage	-30°C / 96H	
3	High Temp. Operating	+70°C / 96H	
4	Low Temp. Operating	-20°C / 96H	
5	High Temperature / Humidity storage	60°C x 90%RH / 96H	
6	Thermal and cold shock	Static state, -20°C (30min) ~70°C (30min), 10 cycles	

9. Packing Method---TBD

- END -

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